

Biology, 2015 EC, entrance exam, version 1

Grade 12 · Biology · 2015 E.C.

Time allowed: 120 minutes

Questions: 98

Total marks: 98

Name _____ Grade _____ ID No. _____ Date _____

1. To be useful in science, a hypothesis must be: (1)
 - A. measurable
 - B. testable
 - C. observable
 - D. correct
2. Which of the following statements about a controlled experiment is true? (1)
 - A. All the variables must be kept the same
 - B. Only one variable is tested at a time
 - C. Everything can be studied by setting up a controlled experiment
 - D. Controlled experiments cannot be performed on living things
3. Which of the following is NOT true a polymer? (1)
 - A. Lipid
 - B. Protein
 - C. Starch
 - D. Nucleic acid
4. The purpose of peer review in science is to ensure that: (1)
 - A. all scientific research is funded
 - B. the results of experiments are correct
 - C. all scientific results are published
 - D. published results meet standards set by the scientific community
5. The enzyme amylase can break glycoside linkages between glucose monomers only if the monomers are in the alpha form. Which of the following could amylase break down? (1)
 - A. Glycogen, starch, and amylopectin
 - B. Glycogen and cellulose
 - C. Cellulose, chitin, and starch
 - D. Starch, amylopectin and cellulose
6. If data from repeated experiments do not support the hypothesis, what is the scientist's next step? (1)
 - A. Give up
 - B. Repeat the experiment
 - C. Revise the hypothesis
 - D. Overturn the theory
7. A hypothesis that is supported many times becomes: (1)
 - A. an experiment
 - B. a theory
 - C. a conclusion
 - D. an observation
8. Which of the following laboratory equipment is used to separate the organelles of the cell according to their density? (1)
 - A. Centrifuge
 - B. Measuring cylinder
 - C. Filter paper with fine pores
 - D. Incubator

-
9. Amino acids are the subunits of: (1)
- | | |
|------------------|------------------|
| A. carbohydrates | B. nucleic acids |
| C. lipids | D. proteins |
-
10. Pitfall traps are used to catch: (1)
- | | |
|----------------------------------|------------------------------|
| A. small ground-dwelling animals | B. small birds in the forest |
| C. flying insects | D. damaged plants |
-
11. Compared to most other substances, a great deal of heat is needed to raise the temperature of water by a given amount. This because water: (1)
- | | |
|----------------------------|-----------------------------|
| A. readily forms solutions | B. has a high heat capacity |
| C. acts as a buffer | D. is an acid |
-
12. The organic molecule tightly bound to the enzyme and required for the activity of enzyme is grouped as: (1)
- | | |
|--------------|---------------------|
| A. apoenzyme | B. activator |
| C. coenzyme | D. prosthetic group |
-
13. Which of the following is an example of a qualitative data? (1)
- A. The temperature decreased from 20°C to 15°C
 - B. The plant's height is 25 centimeters (cm)
 - C. The fish swam in a zigzag motion
 - D. The six pairs of robins hatched an average of three chicks
-
14. The single factor that can be altered in an experiment is the: (1)
- | | |
|-----------------------|-------------------------|
| A. control | B. hypothesis |
| C. dependent variable | D. independent variable |
-
15. Which of the following is NOT an organic molecule found in living organisms? (1)
- | | |
|-----------------|--------------------|
| A. Protein | B. Sodium chloride |
| C. Nucleic acid | D. Lipid |
-
16. Which joins amino acids together? (1)
- | | |
|-------------------------|-------------------|
| A. Peptide bonds | B. Hydrogen bonds |
| C. Van der Waals forces | D. Ionic bonds |
-
17. Which of the following is a hydrophobic material? (1)
- | | |
|---------------|----------|
| A. Table salt | B. Wax |
| C. Sugar | D. Pasta |
-
18. Many mammals control their body temperature by sweating. Which property of water is most directly responsible for the ability of sweat to lower body temperature? (1)
- A. Water's change in density when it condenses
 - B. Water's ability to dissolve molecules in the air
 - C. The release of heat by the formation of hydrogen bonds
 - D. The absorption of heat by the breaking of hydrogen bonds

-
- 19.** Which of the following categories includes all others in the list? (1)
- | | |
|--------------------------|--------------------------|
| A. Monosaccharide | B. Carbohydrate |
| C. Disaccharides | D. Polysaccharide |
-
- 20.** Which of the following statements concerning unsaturated fatty acids is true? (1)
- A.** They are more common in animals than in plants
B. They have double bonds in the carbon chains of their fatty acids
C. They generally solidify at room temperature
D. They contain more hydrogen than do saturated fats having the same number of carbon atoms
-
- 21.** To ensure that a scientific work is free of bias and meet standards set by the scientific community, a research group's work is peer reviewed by: (1)
- | | |
|--|------------------------------------|
| A. anonymous scientific experts | B. the researcher's friends |
| C. the general public | D. lawmakers |
-
- 22.** The structural level of a protein least affected by a disruption in hydrogen bonding is the: (1)
- | | |
|---------------------------|----------------------------|
| A. primary level | B. tertiary level |
| C. secondary level | D. quaternary level |
-
- 23.** Compared to most other substances, a great deal of heat is needed to raise the temperature of water by a given amount. This is because water: (1)
- | | |
|------------------------------------|----------------------------------|
| A. is an acid | B. readily forms solution |
| C. has a high heat capacity | D. acts as a buffer |
-
- 24.** When two amino acids bond: (1)
- A.** a dipeptide is formed through a condensation reaction
B. water is taken in by the product
C. a triglyceride is formed
D. hydrolysis occurs
-
- 25.** Which of the following is not a structural polysaccharide? (1)
- | | |
|---------------------|------------------|
| A. Cellulose | B. Chitin |
| C. Ribose | D. Murein |
-
- 26.** An enzyme speeds up a reaction by: (1)
- | | |
|--------------------------------------|---|
| A. lowering activation energy | B. raising the activation energy |
| C. releasing energy | D. absorbing energy |
-
- 27.** If an enzyme in solution is saturated with substrate, the most effective way to obtain a faster yield of products is to: (1)
- | | |
|-------------------------------------|---------------------------------------|
| A. add more of the enzyme | B. add more substrate |
| C. heat the solution to 90°C | D. add an allosteric inhibitor |
-
- 28.** The rate of an enzyme catalyzed reaction becomes stable after some time due to: (1)
- | | |
|---------------------------------|------------------------------------|
| A. increased temperature | B. increased pH |
| C. saturation of enzymes | D. accumulation of products |

-
- 29.** The nucleus includes all of the following structures EXCEPT: (1)
- | | |
|---------------------|------------------------------|
| A. cytoplasm | B. a nuclear membrane |
| C. DNA | D. a nucleolus |
-
- 30.** If an enzyme is added to a solution where its substrate and product are in equilibrium, what will occur? (1)
- A.** Additional product will be formed
 - B.** Additional substrate will be formed
 - C.** The reaction will change from endergonic to exergonic
 - D.** Nothing; the reaction will stay at equilibrium
-
- 31.** Most CO₂ from catabolism is released during: (1)
- | | |
|------------------------------|-------------------------------------|
| A. Glycolysis | B. The citric acid cycle |
| C. Electron transport | D. Oxidative phosphorylation |
-
- 32.** An enzyme sucrase is an example of: (1)
- | | |
|------------------------|---------------------------|
| A. transferases | B. hydrolases |
| C. lyases | D. oxidoreductases |
-
- 33.** Which two elements are always found in amino acids? (1)
- | | |
|-------------------------------|-----------------------------------|
| A. Nitrogen and sulfur | B. Sulfur and oxygen |
| C. Carbon and oxygen | D. Hydrogen and phosphorus |
-
- 34.** In an enzyme catalyzed reaction, addition of a certain substance to the reaction mixture decreases the reaction. (1)
The substance added must be:
- | | |
|---------------------|---------------------|
| A. substrate | B. coenzyme |
| C. enzyme | D. inhibitor |
-
- 35.** The elements or compounds that enter into a chemical reaction are called: (1)
- | | |
|--------------------|-----------------------|
| A. catalyst | B. substrate |
| C. product | D. active site |
-
- 36.** Which of the following cell types can be rich in lysosomes? (1)
- | | |
|----------------------------|-----------------------------|
| A. Red blood cells | B. White blood cells |
| C. Phagocytic cells | D. Nerve cells |
-
- 37.** Cell membranes consist mainly of: (1)
- | | |
|-------------------------|-------------------------|
| A. lipid bilayer | B. carbohydrates |
| C. protein pumps | D. proteins |
-
- 38.** In a chemical reaction, a reactant binds to an enzyme at a region known as: (1)
- | | |
|--------------------|-----------------------|
| A. catalyst | B. substrate |
| C. product | D. active site |
-
- 39.** The nucleolus: (1)
- A.** produces and assembles the cell's ribosomes
 - B.** stores genetic information
 - C.** is the site where protein is synthesized
 - D.** is the control center of the cell

40. The movement of water molecules across a selectively permeable membrane is known as: (1)

- A. exocytosis
B. endocytosis
C. phagocytosis
D. osmosis

41. Some enzymes are metabolically active in hot springs because: (1)

- A. they are able to maintain a lower internal temperature
- B. high temperatures make catalysis unnecessary
- C. their enzymes have high optimal temperatures
- D. their enzymes are completely insensitive to temperatures

42. Which of the following is an example of active transport? (1)

- A. Facilitated diffusion
B. Diffusion
C. Osmosis
D. Endocytosis

43. The principal chemical compound that living things use to store energy is: (1)

- A. DNA
B. ATP
C. H₂O
D. CO₂

44. Most cells cannot harness heat to perform work because: (1)

- A. heat is not a form of energy
- B. cells do not have much heat; they are relatively cool
- C. temperature is usually uniform throughout a cell
- D. heat can never be used to do work

45. Which of the following is common to plant and animal cells? (1)

- A.** Chloroplast **B.** Central vacuole
C. Centriole **D.** Mitochondria

46. Which of the following is present in a prokaryotic cell? (1)

- A. Mitochondria
B. Nuclear envelope
C. Ribosome
D. Chloroplast

47. The sodium-potassium pump transports: (1)

- A.** both Na⁺ and K⁺ into the cell
- B.** both Na⁺ and K⁺ out of the cell
- C.** Na⁺ out of the cell and K⁺ into the cell
- D.** Na⁺ into the cell and K⁺ out of the cell

48. Which of the following pairs are both organelles concerned with energy transformation? (1)

- A. Mitochondria and nucleus
B. Chloroplast and central vacuole
C. Mitochondria and chloroplast
D. Nucleus and nucleolus

49. Which cell would be best for studying lysosomes? (1)

- A. Muscle cell
B. Bacterial cell
C. Nerve cell
D. Phagocytic white blood cell

50. Which lake zone would be absent in a very shallow lake? (1)

- A.** Benthic zone **B.** Pelagic zone
C. Aphotic zone **D.** Littoral zone

-
- 51.** Materials may be taken into the cell by the following process EXCEPT: (1)
- | | |
|--------------------------|----------------|
| A. exocytosis | B. pinocytosis |
| C. facilitated diffusion | D. endocytosis |
-
- 52.** If the nucleus is removed from a cell, the cell would NOT: (1)
- | | |
|--------------|---------------------------|
| A. reproduce | B. exchange material |
| C. secrete | D. provide its own energy |
-
- 53.** The growth of cells is limited by the ratio between: (1)
- | | |
|--------------------------------|-----------------------------|
| A. organelles and surface area | B. nucleus and cytoplasm |
| C. surface area and volume | D. organelles and cytoplasm |
-
- 54.** Which of the following processes includes all others? (1)
- | | |
|--------------------------|--|
| A. Osmosis | B. Diffusion of a solute across a membrane |
| C. Facilitated diffusion | D. Passive transport |
-
- 55.** In addition to light and chlorophyll, photosynthesis requires: (1)
- | | |
|------------------------------|-----------------------------|
| A. water and oxygen | B. water and sugars |
| C. oxygen and carbon dioxide | D. water and carbon dioxide |
-
- 56.** The first process in light-dependent reactions of photosynthesis is: (1)
- | | |
|----------------------|-----------------------|
| A. light absorption | B. electron transport |
| C. oxygen production | D. ATP formation |
-
- 57.** CAM plants are specialized to survive under what conditions that would harm most other kinds of plants? (1)
- | | |
|------------------------|---------------------|
| A. Low temperature | B. Excess water |
| C. Hot, dry conditions | D. Long day lengths |
-
- 58.** In a typical plant, all of the following factors are necessary for photosynthesis EXCEPT: (1)
- | | |
|----------------|-----------|
| A. chlorophyll | B. oxygen |
| C. light | D. water |
-
- 59.** The net gain of energy from glycolysis is: (1)
- | | |
|--------------------|-----------------------------|
| A. 4 ATP molecules | B. 8 ADP molecules |
| C. 2 ATP molecules | D. 3 pyruvic acid molecules |
-
- 60.** The Krebs cycle takes place within the: (1)
- | | |
|----------------|-----------------|
| A. chloroplast | B. mitochondria |
| C. nucleus | D. cytoplasm |
-
- 61.** Suppose 1000 pea plants are produced from Tt and tt crosses, how many of them are expected to be tall? (1)
- | | |
|--------|--------|
| A. 750 | B. 500 |
| C. 50 | D. 250 |
-
- 62.** The process carried out by yeast that causes bread dough to rise is: (1)
- | | |
|---------------------------|-------------------------|
| A. alcoholic fermentation | B. cellular respiration |
| C. oxygen debt | D. the Krebs cycle |

-
- 63.** During the Krebs cycle: (1)
- A. hydrogen ions and oxygen form water
 - B. the cell releases small amount of energy through fermentation
 - C. each glucose molecule is broken down into two molecules of pyruvic acid
 - D. acetyl is broken down into carbon dioxide in a series of molecules
-
- 64.** The molecular formula for glucose is $C_6H_{12}O_6$. What would be the molecular formula for a polymer made by linking ten glucose molecules together by dehydration reaction? (1)
- A. $C_{60}H_{120}O_{60}$
 - B. $C_{60}H_{102}O_{51}$
 - C. $C_{60}H_{110}O_{51}$
 - D. $C_6H_{12}O_6$
-
- 65.** The light reactions of photosynthesis supply the Calvin cycle with: (1)
- A. light energy
 - B. water and NADPH
 - C. CO_2 and ATP
 - D. ATP and NADPH
-
- 66.** Which metabolic pathway is common to both fermentation and cellular respiration of a glucose molecule? (1)
- A. The citric acid cycle
 - B. The electron transport chain
 - C. Glycolysis
 - D. Synthesis of acetyl CoA from pyruvate
-
- 67.** Some bacteria, when faced with unfavorable environmental conditions, produce structures called: (1)
- A. pili
 - B. endospores
 - C. capsules
 - D. toxins
-
- 68.** During a lytic cycle, after a virus enters the cell, the virus _____. (1)
- A. forms a provirus
 - B. becomes inactive
 - C. replicates
 - D. dies
-
- 69.** _____ is never a part of a virus. (1)
- A. Nucleic acid
 - B. Protein coat
 - C. Viral envelope
 - D. Cell wall
-
- 70.** What does reverse transcriptase do? (1)
- A. It synthesizes protein from RNA
 - B. It synthesizes DNA from RNA
 - C. It synthesizes protein from amino acids
 - D. It synthesizes RNA from DNA
-
- 71.** Which of the following descriptions applies to most protists? (1)
- A. Unicellular prokaryotes
 - B. Unicellular eukaryotes
 - C. Multicellular prokaryotes
 - D. Multicellular eukaryotes
-
- 72.** Which illness is caused by a bacterium? (1)
- A. AIDS
 - B. Diphtheria
 - C. Polio
 - D. Common cold
-
- 73.** What is a capsid? (1)
- A. Viral DNA that inserts into a host's DNA
 - B. A type of plant virus
 - C. A protein coat surrounding a virus
 - D. A rod-shaped bacterium

-
- 74.** What characteristics do viruses share with all living organisms? (1)
- | | |
|-----------------------|-----------------------|
| A. Respiration | B. Replication |
| C. Metabolism | D. Movement |
-
- 75.** Producers in aquatic food chains include: (1)
- | | |
|-----------------------|------------------|
| A. protozoa | B. algae |
| C. slime molds | D. amoeba |
-
- 76.** Most fungi function as _____ in their environment: (1)
- | | |
|---------------------|-----------------------|
| A. consumers | B. decomposers |
| C. producers | D. autotrophs |
-
- 77.** Cell walls of fungi contain: (1)
- | | |
|---------------------|------------------|
| A. cellulose | B. hyphae |
| C. spores | D. chitin |
-
- 78.** If a cell has 12 chromosomes, how many chromosomes will each of its daughter cells have after mitosis and cytokinesis? (1)
- | | |
|--------------|--------------|
| A. 4 | B. 6 |
| C. 12 | D. 24 |
-
- 79.** The feeding relationships among the species in a community determine the community's: (1)
- | | |
|--------------------------------|-----------------------------|
| A. secondary succession | B. species richness |
| C. ecological niche | D. trophic structure |
-
- 80.** Which method is NOT used to protect food against microorganisms? (1)
- | | |
|--------------------|-------------------------|
| A. Heating | B. Sterilization |
| C. Freezing | D. Vaccination |
-
- 81.** The electron transport chain uses the high-energy electrons from the Krebs cycle to: (1)
- | |
|--|
| A. produce glucose |
| B. move H ⁺ ions across the inner mitochondrial membrane |
| C. convert acetyl-CoA to citric acid |
| D. convert glucose to pyruvic acid |
-
- 82.** Prokaryotic cells that have a spherical shape are: (1)
- | | |
|--------------------|-----------------------|
| A. cocci | B. bacilli |
| C. spirilli | D. methanogens |
-
- 83.** Which of the following organisms is incorrectly paired with its trophic level? (1)
- | | |
|---|--|
| A. Cyanobacterium — primary producer | B. Grasshopper — primary consumer |
| C. Zooplankton — primary producer | D. Fungus — detritivore |
-
- 84.** The water in an estuary is: (1)
- | | |
|-----------------------------|---|
| A. salt water only | B. fresh water only |
| C. poor in nutrients | D. a mixture of fresh water and salt water |

-
- 85.** An increase in the greenhouse effect causes an increase in: (1)
- | | |
|--------------------------|------------------|
| A. carbon dioxide | B. oxygen |
| C. temperature | D. water |
-
- 86.** Permafrost characterizes the biome called: (1)
- | | |
|-------------------------|-------------------|
| A. taiga | B. savanna |
| C. boreal forest | D. tundra |
-
- 87.** The maximum number of organisms of a particular species that can be supported by an environment is called: (1)
- | | |
|-----------------------------|------------------------------|
| A. logistic growth | B. exponential growth |
| C. carrying capacity | D. population density |
-
- 88.** The demographic transition is considered complete when: (1)
- A.** population growth stops
 - B.** the birth rate is greater than death rate
 - C.** the death rate begins to fall
 - D.** the death rate is greater than the birth rate
-
- 89.** In the presence of unlimited resources and in the absence of disease and predation, what would probably happen to a bacterial population? (1)
- | | |
|------------------------------|------------------------|
| A. logistic growth | B. endangerment |
| C. exponential growth | D. extinction |
-
- 90.** If the number of female population is greater than the male population in a given country, its human population tends to: (1)
- | | |
|---------------------|---------------------------|
| A. disappear | B. increase |
| C. decrease | D. remain constant |
-
- 91.** Which of the following aquatic biomes have many strata like that of equatorial rain forest? (1)
- | | |
|------------------------|----------------------|
| A. Wetland | B. Estuarine |
| C. Pelagic zone | D. Coral reef |
-
- 92.** A plant has a genotype AaBb. If there is no linkage of genes, the gametes it produces are: (1)
- | | |
|---------------------|------------------------------|
| A. AB and ab | B. AA, aa, BB, and Bb |
| C. Aa and Bb | D. AB, Ab, aB, and ab |
-
- 93.** Fresh water biome includes: (1)
- | | |
|---|---|
| A. ponds, lakes, rivers and wetlands | B. estuaries, ponds, lakes, and rivers |
| C. wetlands, estuaries and rivers | D. pelagic zone, wetlands, and estuaries |
-
- 94.** Before DNA could definitely be shown to be genetic material in cells, scientists had to show that it could: (1)
- A.** tolerate high temperature
 - B.** carry and make copies of information
 - C.** be taken down into small sub units
 - D.** be modified in response to environmental conditions

-
- 95.** A limiting factor that depends on a population size is called: (1)
- A. density-dependent limiting factor B. density-independent limiting factor
C. predator-prey relationship D. parasitic relationship
-
- 96.** Two parents of genotype Aa are cross-bred. The alleles show complete dominance. What proportion of the offspring will genotypically look like their parents? (1)
- A. 1/2 B. 3/4
C. 1/4 D. 2/3
-
- 97.** Tropical rain forest is characterized by: (1)
- A. low precipitation and low temperature B. low precipitation and high temperature
C. high precipitation and low temperature D. high precipitation and high temperature
-
- 98.** Group of different species that live together in a defined area make up a (an): (1)
- A. population B. ecosystem
C. community D. biosphere

— END OF EXAMINATION —

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